USGLoc: Uncertainty-Aware Semantic Visual Localization from 3D Gaussian Splatting

***Abstract*—A visual localizer should tell a robot where the camera is and how much to trust that answer, because the pose is fused into a Bayesian state estimator. Recent semantic 3D Gaussian Splatting (3DGS) localizers are accurate but return a single pose with no confidence, while uncertainty-aware mapping methods quantify map reliability but do not localize. We present USGLoc, an evidential localizer built on one principle: the map's own uncertainty drives every stage and survives to the output. Each primitive of an evidential semantic Gaussian map carries a Dempster–Shafer belief over open-vocabulary semantics and an explicit vacuity mass, and the query is encoded into the same space. Vacuity gates 2D–3D semantic matching, which turns retrieval ambiguity into a pose mixture, gates the correspondences of a render-and-match correction, and weights the rendering residual per pixel. A tempered Laplace approximation turns each refined hypothesis into a covariance, giving a Gaussian-mixture posterior on $SE(3)$ that condenses to one Gaussian for a filter. On 7Scenes, USGLoc localizes prior-free to 0.82 cm/0.27° on average, and its uncertainty orders its own errors far better than the confidence signal of a strong scene-coordinate baseline (sparsification error 0.063 against 0.166). The tempered covariance is close to nominal at the 68% level and over-confident in the tails, a residual we trace to multi-modal scenes.**

***Index Terms*—Localization, 3D gaussian splatting, semantic scene understanding, uncertainty quantification.**

I. Introduction

VISUAL localization estimates the 6-DoF camera pose of a query image inside a previously mapped scene, which is a core capability for indoor mobile robots and micro aerial vehicles (MAVs) [1]–[3]. On these platforms the pose is rarely consumed on its own. It enters a recursive Bayesian estimator, such as an EKF or UKF, and is fused with inertial and other cues [4]. This fusion is only as sound as the *uncertainty* attached to each measurement. A pose without a covariance forces the estimator to assume a fixed noise model. A fixed model cannot reflect scene ambiguity, occlusion, dynamics, or texture sparsity. These are exactly the conditions under which localization degrades. A localizer should therefore output a *calibrated distribution* over pose, not a point.

Classical localization is dominated by feature-based and regression-based methods. Feature-based pipelines match 2D keypoints to a 3D model and solve Perspective-n-Point with RANSAC [5]. Absolute pose regression (APR) [6], [7] and scene coordinate regression (SCR) [8], [9] learn the mapping end-to-end. 3D Gaussian Splatting (3DGS) [1] then brought high-fidelity novel-view synthesis at real-time rates. It has become an attractive map representation, and a family of 3DGS localizers now refines pose by inverting or matching against the rendered map [10]–[13]. Semantics improve robustness further. SGLoc [2] retrieves a coarse pose by aligning 2D and 3DGS scene-level semantic descriptors. Chu and Kim [3] use a language model to select "stable" objects for a robust submap. These methods are accurate, but they share two limitations. First, they emit a single pose and *say nothing about its reliability*. Second, they handle scene dynamics implicitly or through hard, discrete object selection. Discrete selection cannot express graded confidence.

Uncertainty-aware methods stop short of this. Measurement-error distributions have been learned for 2D retrieval localization [14], particle localizers over radiance fields report only a MAP pose [15], and UGS-Loc [16] uses Fisher-information uncertainty only internally before emitting a point estimate. Evidential semantic mapping [17], [18] propagates per-observation uncertainty with Evidential Deep Learning (EDL) [19] and Dempster–Shafer theory (DST) [20], but that uncertainty never reaches the pose a robot consumes. To our knowledge, no method outputs a calibrated pose distribution from a Gaussian-splatting map.

USGLoc fills that gap with one principle: *the map's own uncertainty should drive every stage of localization and survive to the output*. Each object-level primitive of the map stores geometry, a CLIP [21] embedding and a DST belief with an explicit vacuity mass; the query is segmented [22] and encoded by the *same* evidential head, so map and query confidences share one scale. Vacuity then gates 2D–3D semantic matching, turning retrieval ambiguity into a weighted pose mixture, gates the correspondences of a render-and-match correction, and, alpha-blended through the rasterizer like color, weights the refinement per pixel. At the optimum a tempered Laplace approximation yields a per-hypothesis covariance, and the mixture condenses to one Gaussian for a filter. Our contributions are:

- **Evidential localization on a 3DGS map.** We introduce the first localizer that outputs a *calibrated* $SE(3)$ *pose distribution* from a Gaussian-splatting map.
- **One propagated uncertainty channel.** Conflict-aware DST fusion builds the map; the same vacuity gates semantic matching and the correction's correspondences and weights the refinement per pixel, replacing discrete robust-object selection with graded, evidence-driven trust.
- **Calibrated covariance from rendering.** A Laplace approximation at the refinement optimum converts the weighted photometric objective into a per-hypothesis covariance at negligible cost. A single temperature, fitted on a held-out split, makes it honest.

II. Related Work

A. Visual Localization

APR methods directly regress pose from an image [6], [7]. They are fast and storage-light. However, they tie accuracy

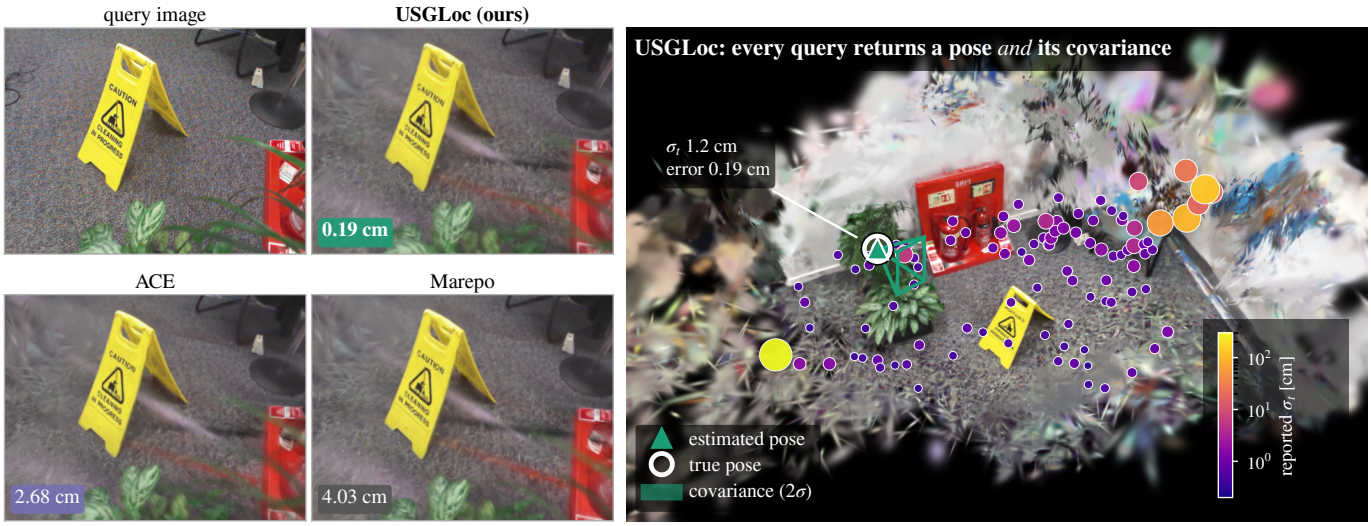


Fig. 1. USGLoc returns a pose *and* its covariance. Left: a 7-Scenes *fire* query and the Gaussian map rendered at the pose each method returned, with translation error. Right: the same map from a free camera, every test query coloured by the σ_t we reported for it, with one query showing its estimated pose, the true pose and the 2σ covariance. A favourable query, chosen for legibility; Tables I–II give the distribution.

to the training distribution. They also lack explicit geometric reasoning. SCR methods regress 2D–3D correspondences and recover pose by PnP+RANSAC [8], [9]. Geometry improves their accuracy. However, they generalize poorly across scenes and degrade in cluttered indoor layouts. Recent extensions add per-coordinate uncertainty to prefilter correspondences [23]. Feature-based pipelines [5] remain strong. However, they rely on low-level features that are sensitive to lighting and texture. Some of these methods do estimate uncertainty. In every case, that uncertainty is consumed internally. None reports a calibrated pose covariance.

B. Rendering-Based Localization

NeRF-based methods refine pose by synthesizing views but render slowly; particle variants [15] keep a nonparametric belief yet report only its MAP point. 3DGS [1] renders fast from a compact map, and 3DGS localization has expanded rapidly: iComMa [10] inverts the renderer by gradient descent, and GS-CPR [12], GSplatLoc [11], STDLoc [13] and SplatHLoc [24] match and refine ever more accurately against the rendered map. UGS-Loc [16] is the most related: it refines an SCR prior with Monte Carlo pose particles weighted by a rendered Fisher-information uncertainty map, which also biases PnP-RANSAC sampling. Its uncertainty is post-hoc, purely geometric and consumed internally; the output is a single pose without covariance, calibration or semantics. SGLoc [2] adds semantic 2D-to-3DGS retrieval for prior-free coarse pose, and Chu and Kim [3] mask dynamic content with LLM-selected robust objects; their gating is discrete, ours is continuous and driven by the map's own evidence. All of them output point poses. USGLoc keeps their accuracy mechanisms and adds propagated, graded uncertainty and a calibrated pose distribution.

C. Uncertainty in Localization and Semantic Mapping

Bayesian and ambiguity-aware pose regressors output uncertainty but need static scenes and dense pose labels. Evidential learning has reached scene-coordinate regression with NeRF-derived uncertainty [25], without a metric map at query time. Wu and Campbell [14] learn a 2D measurement-error distribution and show it is best modeled as a Gaussian mixture under adverse conditions; we adopt this distributional view and realize it over $SE(3)$. In semantic mapping, EBS [17] and E^2-BKI [18] model semantic uncertainty with EDL [19] and DST [20], E^2-BKI with anisotropic Gaussian primitives, and radiance-field uncertainty is studied through Fisher information [26]. In all of these the uncertainty stays inside mapping or tracking. We transplant the evidential machinery to localization: where mapping uses it to score observations, our localizer uses it to score correspondences and pixels.

III. Method

Fig. 2 and Algorithm 1 summarize USGLoc. We first describe the evidential semantic representation shared by the map and the query. We then describe the two-stage coarse-to-fine localizer that turns this representation into a calibrated pose distribution.

Let $\mathcal{M} = \{\mathcal{G}_i\}_{i=1}^{M}$ be a pre-built semantic Gaussian map and I_q a query RGB(-D) image. Prior localizers seek a point estimate $\hat{T} = \arg\max_T s(I_q, T; \mathcal{M})$ for some scoring function s and a pose $T = [R\,|\,t] \in SE(3)$. We instead estimate the full posterior

$$p\big(T \,\big|\, I_q, \mathcal{M}\big), \qquad T \in SE(3), \tag{1}$$

and require it to be *calibrated*: a credibility-α region of (1) should contain the ground-truth pose with empirical frequency α. To make uncertainty first-class on both sides, we represent each modality as a set of *evidential semantic instances*. A reference primitive is

$$\mathcal{G}_i = \big(\boldsymbol{\mu}_i,\ \boldsymbol{\Sigma}_i,\ f_i,\ \mathbf{p}_i,\ u_i\big), \tag{2}$$

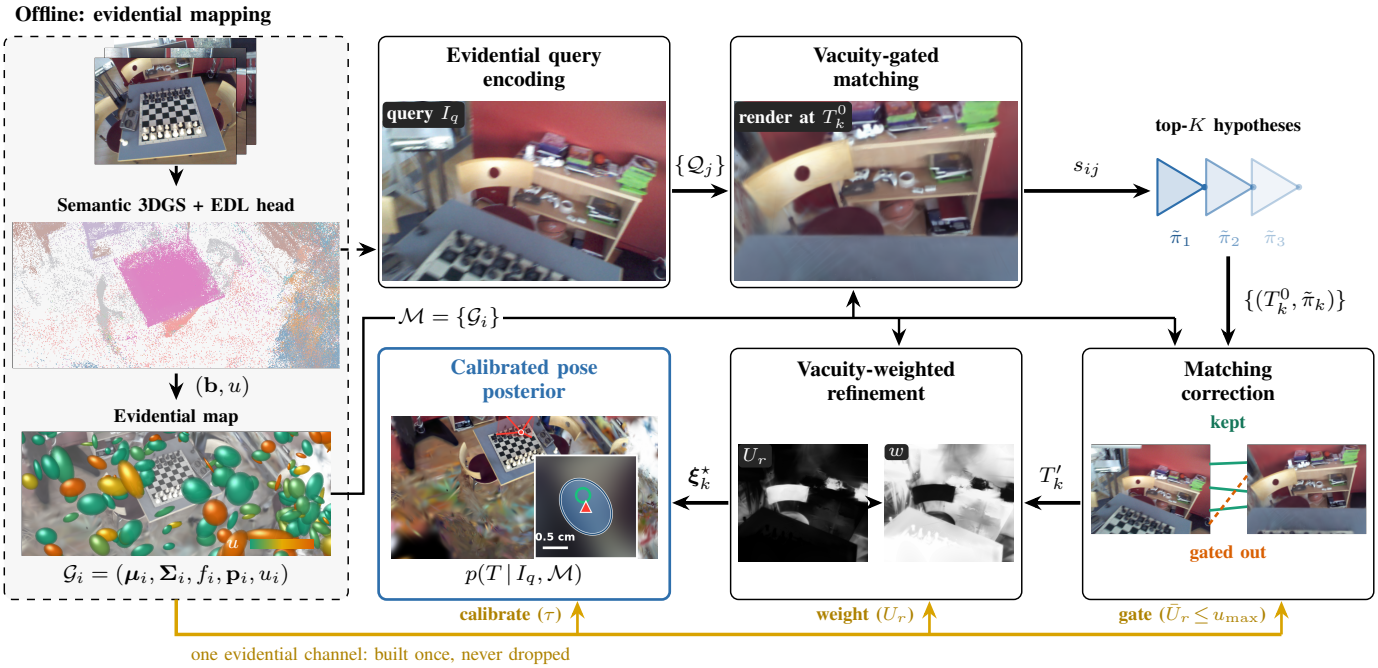


Fig. 2. USGLoc on one Chess query. **Offline** (left): a semantic 3DGS map whose EDL head gives every Gaussian an opinion $(\mathbf{b}, u)$ is fused into evidential primitives $\mathcal{G}_i$ (ellipsoids coloured by vacuity u_i). **Coarse** (top): the query is encoded by the same head and matched with an evidential gate into top-K hypotheses. **Fine** (bottom, right to left): gated matching correction, vacuity-weighted refinement ($w = e^{-\gamma U_r}$) and a tempered Laplace posterior (inset: 2σ ellipse around the true pose). The same vacuity (gold) gates, weights and calibrates.

with mean $\boldsymbol{\mu}_i \in \mathbb{R}^3$, covariance $\boldsymbol{\Sigma}_i \in \mathbb{R}^{3\times3}$ (anisotropic geometry), CLIP embedding $f_i \in \mathbb{R}^d$, semantic probability $\mathbf{p}_i \in \Delta^{C-1}$ over C categories, and a scalar vacuity (uncertainty) mass $u_i \in [0,1]$, the conflict-aware vacuity of the fused primitive (Sec. III-A). A query instance $\mathcal{Q}_j = (\mathbf{x}_j, a_j, f_j^q, \mathbf{p}_j^q, u_j^q)$ stores its 2D centroid $\mathbf{x}_j$, area a_j, and the same semantic triplet.

A. Evidential Semantic Gaussian Map

Evidential semantics. Consider a semantic segmentation network with an evidential head [19]. Its non-negative outputs are interpreted as evidence $e = [e_1, \dots, e_C]$ for a Dirichlet density with parameters $\alpha_c = e_c + 1$. With $S = \sum_{c=1}^{C} \alpha_c$, the expected class probability, belief mass, and vacuity are

$$\hat{p}_c = \frac{\alpha_c}{S}, \qquad b_c = \frac{\alpha_c - 1}{S}, \qquad u = \frac{C}{S}, \tag{3}$$

satisfying $u + \sum_c b_c = 1$. Vacuity u is large when total evidence S is small, i.e., when the network is uncertain. This is the per-observation uncertainty we will propagate.

From Gaussians to evidential primitives. We build the map with an open-vocabulary semantic 3DGS representation that assigns an identity/semantic feature to each Gaussian [27]. Gaussians are clustered by spatial and semantic affinity into object-level primitives (as in [3]). For cluster $\mathcal{C}_i$, geometry is summarized by the empirical mean $\boldsymbol{\mu}_i$ and covariance $\boldsymbol{\Sigma}_i$ of member centroids. The CLIP embedding f_i is a vacuity-weighted average of member embeddings. The semantic belief is obtained by DST-fusing the member beliefs. For two belief–vacuity pairs $(\mathbf{b}_1, u_1), (\mathbf{b}_2, u_2)$, Dempster's rule gives [18]

$$b^c = \tfrac{1}{1-\eta}\big(b_1^c b_2^c + b_1^c u_2 + b_2^c u_1\big), \quad u = \tfrac{1}{1-\eta}\, u_1 u_2, \tag{4}$$

where $\eta = \sum_{x \neq y} b_1^x b_2^y$ measures conflict. Applying (4) pairwise over $\mathcal{C}_i$ yields $(\mathbf{b}_i, u_i)$. This pair is converted back to $\mathbf{p}_i$ via $p_i^c = b_i^c + u_i/C$. Fusing within a cluster suppresses isolated mispredictions. Dempster's normalization, however, *renormalizes conflict away*: two confident but contradictory member opinions, the signature of an object that moved between mapping passes, fuse to a deceptively low vacuity (Zadeh's paradox). The reported maps therefore use Yager's variant [28] of (4), which assigns the conflict mass to vacuity instead; adding the mean pairwise conflict $\bar{\eta}_i$ to Dempster's vacuity, $u_i \leftarrow \min(1, u_i + \bar{\eta}_i)$, is an equivalent option. The primitive vacuity thus reflects both lack of evidence and persistent disagreement. The map is the set $\mathcal{M} = \{\mathcal{G}_i\}$ of (2). For retrieval, it is partitioned into overlapping submaps sampled along the mapping trajectory [2].

B. Evidential Query Encoding

Given I_q, we obtain instance masks with Fast-SAM [22] and crop each instance. Each crop passes through CLIP [21] to obtain f_j^q. It also passes through the same evidential head as the map to obtain $(\mathbf{p}_j^q, u_j^q)$ via (3). Per-instance geometry (normalized centroid $\mathbf{x}_j \in [0,1]^2$ and area a_j) is encoded by small MLPs. A query instance descriptor is

$$d_j^q = \mathrm{MLP}\big[\,\phi(f_j^q) \,\|\, \psi_{\mathbf{x}}(\mathbf{x}_j) \,\|\, \psi_a(a_j)\,\big], \tag{5}$$

with ϕ a projection of the CLIP feature into the shared space and $\|$ concatenation. Reference descriptors d_i^G are formed analogously from $(f_i, \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$ projected into the camera frame. Because both modalities pass through the same evidential head, u^q and u^G are directly comparable.

C. Uncertainty-Weighted Matching and the Pose Mixture

Gated instance similarity. We score a cross-modal instance pair by appearance similarity modulated by an evidential gate. The gate suppresses pairs in which either side is uncertain:

$$s_{ij} = \underbrace{\cos\big(d_i^G,\, d_j^q\big)}_{\text{semantic similarity}} \cdot \underbrace{\exp\big(-\lambda_u\,(u_i^G + u_j^q)\big)}_{\text{evidential gate}}, \tag{6}$$

where $\lambda_u \geq 0$ controls gate sharpness. This is the localization analogue of the uncertainty-aware kernel in E^2-BKI [18]. Confident–confident matches dominate. High-vacuity instances contribute only weakly.

Scene descriptors and retrieval. We follow the contrastive retrieval design of SGLoc [2]. Instance descriptors are aggregated into a scene-level descriptor by multi-head self-attention with a feed-forward layer. From $\hat{d} = Q + \mathrm{Attn}(Q, K, V)$ and $\hat{D} = \hat{d} + \mathrm{FFN}(\hat{d})$, we obtain attention weights $W = \mathrm{softmax}(\mathrm{MLP}(\hat{D}))$ and the scene descriptor $F = \sum_j \hat{D}_j W_j$ for each modality. Cosine similarity between the query scene descriptor F^q and each submap descriptor F_k^G ranks submaps. The top-k submaps provide k coarse pose hypotheses with retrieval-score weights $\tilde{\pi}_k \propto \mathrm{sim}(F^q, F_k^G)$. The gate (6) is injected into the instance-level term of the scene aggregation. Retrieval therefore inherits the same uncertainty weighting. Ambiguous scenes return several comparably scored submaps. The hypotheses therefore form a *pose mixture*. This mixture is the $SE(3)$ realization of the non-Gaussian error observed in [14].

Training objective. The retrieval encoders are trained with a symmetric InfoNCE loss that aligns 2D and 3D scene descriptors [2]. The semantic heads are trained with the evidential loss of [19]: the Bayes risk of the Brier score under the Dirichlet plus a KL term that drives spurious evidence toward the uniform prior,

$$\mathcal{L}_{\mathrm{EDL}} = \sum_c \Big[(y_c - \hat{p}_c)^2 + \tfrac{\hat{p}_c(1-\hat{p}_c)}{S+1}\Big] + \lambda_t\,\mathrm{KL}\big[D(\tilde{\boldsymbol{\alpha}})\,\|\,D(\mathbf{1})\big], \tag{7}$$

with $\tilde{\boldsymbol{\alpha}}$ the evidence of misleading classes. The correction and refinement stages (below) are training-free.

D. Evidential Matching Correction

Retrieval initializes each hypothesis at an anchor pose of the map. Held-out queries lie about 13–15 cm from the nearest anchor, while photometric refinement (Sec. III-E) converges only within 10–35 cm. We close this gap with a geometric correction stage. The map is rendered at the hypothesis pose T. This yields the RGB render I_r, the expected depth D_r, and the vacuity map U_r of (9) in one rasterization pass. A frozen dense matcher (LoFTR [29]) returns correspondences $(\mathbf{x}_k^q, \mathbf{x}_k^r)$ between query and render. Each render-side keypoint is lifted to a world point through the rendered depth, $\mathbf{X}_k = T^{-1}\pi^{-1}\big(\mathbf{x}_k^r, D_r(\mathbf{x}_k^r)\big)$. The corrected pose solves

$$T' = \arg\min_T \sum_k \mathbb{1}\big[\bar{U}_r(\mathbf{x}_k^r) \leq u_{\max}\big]\,\rho\big(\big\|\pi(T\mathbf{X}_k) - \mathbf{x}_k^q\big\|\big), \tag{8}$$

by PnP-RANSAC with Levenberg–Marquardt refinement on the inliers, iterated (re-render, re-match) for five rounds, keeping the round with the most inliers. Here $\bar{U}_r = U_r/\alpha_r$ is the alpha-normalized rendered vacuity: the blended channel of (9) is bounded by the render alpha α_r, so the normalization makes the gate act on content, not on coverage. Matches near depth discontinuities are also removed, since expected depth blends foreground and background there. The indicator is the correspondence form of the evidential gate: the same channel that gates instances in (6) and weights pixels in (10) here removes matches on vacuous content such as unobserved regions, floaters, and ambiguous geometry. The correction basin is set by the matcher's view overlap, not by the photometric loss surface. Measured on chess, it recovers initializations 30 cm and 15° away on every frame tested, and up to 60 cm and 25° on most. It therefore rescues hypotheses that photometric refinement cannot reach, and near the optimum it polishes below the rendering-resolution noise floor. Snapped hypotheses are re-ranked by render fidelity; the inlier count enters only the mixture weights (13). When too few gated matches survive, or when the corrected pose does not improve render fidelity, the hypothesis keeps its anchor pose and falls back to the coarse-to-fine photometric path.

E. Uncertainty-Aware Refinement and Pose Covariance

Weighted rendering objective. Each corrected hypothesis T' is refined by differentiable rendering [10]. We parameterize the update in the tangent space as $T(\boldsymbol{\xi}) = \exp(\boldsymbol{\xi}^\wedge)\,T'$ with $\boldsymbol{\xi} \in \mathfrak{se}(3)$. We render both an RGB image $I_r(\boldsymbol{\xi})$ and a per-pixel *uncertainty map* U_r. The map U_r is obtained by alpha-blending primitive vacuity exactly as color/depth are blended in 3DGS:

$$U_r(\mathbf{x}) = \sum_i u_i\,\alpha_i(\mathbf{x}) \prod_{j<i}\big(1 - \alpha_j(\mathbf{x})\big), \tag{9}$$

where $\alpha_i(\mathbf{x})$ is the opacity contribution of primitive i at pixel $\mathbf{x}$. The refinement minimizes the per-pixel weighted photometric term

$$E(\boldsymbol{\xi}) = \sum_{\mathbf{x}} w(\mathbf{x})\,\big\|I_q(\mathbf{x}) - I_r(\mathbf{x};\boldsymbol{\xi})\big\|^2, \tag{10}$$

with $w(\mathbf{x}) = \exp\big(-\gamma\,U_r(\mathbf{x})\big)$. Pixels whose supporting Gaussians carry little evidence—rarely observed, texture-poor or semantically ambiguous regions—receive small weight *continuously*, rather than through the discrete mask of [3]. The weight acts on the map's own evidence, not on map–query disagreement: content that changed after mapping still renders confidently, and in a controlled test (deleting object instances from a trained map) the weighting gave no robustness to staleness. Its benefit is on well-mapped but low-evidence content. The weights are held fixed at each iterate (IRLS-style). Gradients flow through the photometric residual only. If w were differentiated through, the optimizer could reduce (10) by steering high-residual pixels into high-vacuity regions instead of correcting the pose.

Covariance via Laplace approximation. At the optimizer's solution $\boldsymbol{\xi}^\star$, let $\mathbf{r}$ be the stacked weighted residuals of (10) and $J = \partial\mathbf{r}/\partial\boldsymbol{\xi}|_{\boldsymbol{\xi}^\star}$ the Jacobian. A Laplace approximation of the per-hypothesis posterior is a tangent-space Gaussian

$$\boldsymbol{\xi} \sim \mathcal{N}\big(\boldsymbol{\xi}^\star,\ \boldsymbol{\Sigma}_{\boldsymbol{\xi}}\big), \qquad \boldsymbol{\Sigma}_{\boldsymbol{\xi}} \approx \hat{\sigma}^2\,\big(J^\top W J\big)^{-1}, \tag{11}$$

where $W = \mathrm{diag}\{w(\mathbf{x})\}$ and $\hat{\sigma}^2$ is estimated from the residuals. After refinement, one more correction round (8) is applied at the optimum and accepted only when render fidelity improves. A short refinement then re-establishes stationarity of (10) at the accepted pose, so (11) is evaluated at an actual optimum. A flat optimum (poorly constrained pose) yields a large covariance. A sharp basin yields a tight one. The reported uncertainty therefore reflects the actual conditioning of the alignment. Real photometric residuals are heavy-tailed and spatially correlated. The raw Laplace covariance can therefore remain optimistic. We apply a single scalar inflation factor calibrated on a held-out split (a temperature on $\boldsymbol{\Sigma}_{\boldsymbol{\xi}}$). Sec. IV-C compares raw and tempered covariances.

Pose distribution and condensation. A PSNR/feature-similarity check [2] prunes hypotheses whose renders diverge from I_q. The surviving hypotheses form a Gaussian mixture on $SE(3)$, with each component expressed in the tangent space at its mode:

$$p\big(T \mid I_q, \mathcal{M}\big) = \sum_k \pi_k \, \mathcal{N}\big(\boldsymbol{\xi};\, \boldsymbol{\xi}_k^{\star},\, \boldsymbol{\Sigma}_{\boldsymbol{\xi},k}\big), \tag{12}$$

The weights must score *where each hypothesis ended*, not where it began. The retrieval score $\tilde{\pi}_k$ describes the anchor a hypothesis was initialized from, and the correction then often moves it by tens of centimeters, so on a repetitive scene $\tilde{\pi}_k$ carries no information about which mode is correct while still dominating any product it enters. We therefore form

$$\pi_k \;\propto\; s_k \, \max\big(n_k / \max_j n_j,\; \epsilon\big), \tag{13}$$

where n_k is the inlier count of the correction (8) at hypothesis k, ϵ keeps a component that failed to register from being zeroed outright, and s_k is the *anchor agreement*: the number of shortlisted anchors whose corrected poses coincided (within 1 cm and 2°) and were merged into hypothesis k. A pose that several anchors' corrections converge to is corroborated; a registration on the wrong tread is usually specific to the anchor it started from. The factor was adopted on train-side validation frames. Render fidelity is a weaker discriminator, since two poses one repeated element apart render almost equally well, so it prunes hypotheses but does not weight them. Table IV quantifies this: on Stairs the median error is 5.92 cm when components are weighted by the retrieval prior and 2.83 cm by inlier count.

Point estimate and condensation. For a downstream Bayesian filter we condense (12) by moment matching [14]: $\boldsymbol{\mu} = \sum_k \pi_k \boldsymbol{\xi}_k^{\star}$ and $\boldsymbol{\Sigma} = \sum_k \pi_k \big(\boldsymbol{\Sigma}_{\boldsymbol{\xi},k} + (\boldsymbol{\xi}_k^{\star} - \boldsymbol{\mu})(\boldsymbol{\xi}_k^{\star} - \boldsymbol{\mu})^{\top}\big)$, which is transported to the inertial frame as the measurement covariance for an EKF/UKF [4]. The moment-matched *mean*, however, is a poor point estimate when the posterior is genuinely multi-modal: it lies between the modes, at a pose the model itself considers unlikely. We therefore report the highest-weight mode as the pose and retain $\boldsymbol{\Sigma}$, which includes the between-component spread and so widens exactly when the modes disagree. The distinction is immaterial on unimodal scenes and substantial on Stairs, where the modes sit one stair tread apart.

Algorithm 1 USGLoc inference for one query image

Require: I_q, map $\mathcal{M}$, anchors $\{T_a, F_a^G\}$, temperature τ
Ensure: $p(T \mid I_q, \mathcal{M})$ (12), pose $\hat{T}$, covariance $\boldsymbol{\Sigma}$
Evidential query encoding
$\{\mathcal{Q}_j\} \leftarrow$ Fast-SAM, CLIP and EDL (3) per instance
$d_j^q \leftarrow$ (5); $F^q \leftarrow$ attention pooling of $\{d_j^q\}$
Vacuity-gated matching
rank anchors by s_{ij} (6) and $\cos(F^q, F_a^G)$ *gate*
$\{(T_k^0, \tilde{\pi}_k)\}_{k=1}^{K'} \leftarrow$ the top-K' anchors
Matching correction
for $k = 1, \ldots, K'$ **do**
  $(T_k', n_k) \leftarrow$ (8) from T_k^0, $\bar{U}_r \le u_{\max}$ *gate*
  if unregistered: $T_k' \leftarrow$ coarse-to-fine (10)
end for
merge equal T_k' ($s_k \leftarrow$ count); keep the top K
Vacuity-weighted refinement
for each kept k **do**
  $\boldsymbol{\xi}_k^{\star} \leftarrow \arg\min E(\boldsymbol{\xi})$ (10), $w = e^{-\gamma U_r}$ *weight*
  re-correct at $\boldsymbol{\xi}_k^{\star}$ if fidelity improves
  $\boldsymbol{\Sigma}_{\boldsymbol{\xi},k} \leftarrow \tau\, \hat{\sigma}^2 (J^{\top} W J)^{-1}$ (11) *calibrate*
end for
Calibrated pose posterior
prune; $\pi_k \propto s_k \max(n_k / \max_j n_j, \epsilon)$ (13)
$\hat{T} \leftarrow$ top mode; $\boldsymbol{\Sigma} \leftarrow$ moment match of (12)
return $p(T \mid I_q, \mathcal{M})$, $\hat{T}$, $\boldsymbol{\Sigma}$

IV. Experiments

A. Experimental Setup

Datasets. We evaluate on 7Scenes [30], the standard indoor RGB-D relocalization benchmark, and report 12Scenes [31] results in Table II. On both we adopt the *SfM pseudo-ground truth* and its per-scene SfM focal lengths [32]. This is the convention used by all recent 3DGS localizers [12], [13], [16], [24]. The two references differ by about 1 cm on the same frame, which exceeds the gaps between recent methods, so DSLAM-referenced numbers are not directly comparable. One example is SGLoc [2], which also evaluates only four of the seven scenes; the supplementary material documents why a render-based localizer cannot be evaluated under its protocol.

Baselines. APR: PoseNet [6], Marepo [7]. SCR: ACE [9], DSAC* [8]. 3DGS-based: GSplatLoc [11]. For calibration, no baseline reports a covariance. We therefore compare against ACE's inlier count (Fig. 4), against a softmax (non-evidential) variant of our method on matched maps (Table IV), against a fixed-covariance point estimator in the fusion study, and against the ideal χ^2 reference.

Metrics. Accuracy: median translation (cm) and rotation (°) error, and success rate at $5\,\mathrm{cm}/5°$ and $2\,\mathrm{cm}/2°$. *Calibration* is our focus. We report three metrics. (i) *Coverage* at the $68.3/95.4/99.7\%$ credibility levels. A query is covered when the Mahalanobis error $d^2 = \mathbf{e}^{\top}\boldsymbol{\Sigma}^{-1}\mathbf{e}$ falls inside the corresponding χ^2 quantile. We compare against the ideal χ^2 curve as in [14]. (ii) AUSE (sparsification error) [33] for translation and rotation. This tests whether ranking predictions by predicted uncertainty and discarding the worst monotonically reduces error. (iii) Expected calibration error (ECE) [34] over

the credibility grid. AUSE integrates the sparsification gap over the first 90% of removed queries. Runtime is reported in seconds per query.

Implementation. Each map is fitted with gsplat from back-projected depth, with an L1 term on the sensor depth (weight 0.2) that suppresses floaters (7,000 iterations at 320×240 on every 10th mapping frame); FastSAM-s masks with CLIP ViT-B/16 crop embeddings are distilled onto the Gaussians (every 20th frame), and the shared evidential head converts the distilled features into opinions with observation-scaled evidence ($n_0{=}4$; an unobserved Gaussian is fully vacuous). The fit resolution is chosen per scene on train-side frames, and the selected map is fine-tuned at 640×480 for 5,000 iterations with densification off, so the distilled semantics stay valid (supplementary material). Primitives are voxel (0.4 m) $\times$ class groups; anchors are every 20th mapping frame. The head ($C{=}25$ indoor classes, a two-layer MLP on CLIP features) is trained once per benchmark on CLIP pseudo-labels and shared by map and query; the retrieval encoders use symmetric InfoNCE. A CLIP whole-image shortlist (top-10 anchors) is reranked by the gated similarity; the top-8 hypotheses are corrected (frozen LoFTR at 640×480, $u_{\max}{=}0.9$, PnP-RANSAC at 2 px with LM, five rounds), deduplicated and re-ranked by render fidelity, and the best $K{=}3$ receive 60 refinement iterations and one fidelity-checked polish round; hypotheses the matcher cannot snap fall back to a coarse-to-fine pyramid ($160\times120 \rightarrow 640\times480$). We use $\lambda_u{=}2$ and $\gamma{=}3$, tuned with the other continuous hyperparameters on a validation split of train-side frames; the shortlist, the pyramid and the fidelity check were adopted from measurements on the Chess and Stairs test splits. The temperature is fitted on 30 train-side frames with $\tau = \mathrm{med}(d^2)/\chi^2_{6,0.5}$ and never on test queries. Each scene is evaluated once on 100 uniformly subsampled test frames (700 queries).

B. Localization Accuracy

Table I reports median errors on 7Scenes. USGLoc is sub-centimetre on four of seven scenes, has the lowest average error in the table (0.82 cm/0.27°) and the highest success rate at 2 cm/2° (87%), and is best in translation on every scene except Kitchen (level with GSplatLoc on Chess); Fig. 5 shows our best cases against ACE and Marepo, and Fig. 6 the stages. On our earlier maps, fitted without the depth term, the frozen LoFTR matcher throughout instead of the MASt3R→field cascade averaged 1.05 cm/0.33° against 0.90 cm/0.30°, with success rates at 5 cm/5° of 85 and 88%. We do not claim state-of-the-art point accuracy: methods that refine an external scene-coordinate prior report lower errors [12]. The failure tail remains our main weakness: 3.6% of our queries exceed 5 cm (13% on Pumpkin), against 2.9% for ACE on the same frames.

Accuracy is bounded by the map. Correcting from the ground-truth pose, which removes retrieval and selection, still leaves a *correction floor* of 0.83 cm on average with the LoFTR matcher, and four of seven scenes report within 0.05 cm of their floor (supplementary material). Stairs is the exception: a self-similar staircase yields confident correspondences on the *wrong* tread, so the pose is wrong while the inlier count is high, which is the regime the pose mixture exists to express. Seeded from an external pose prior instead of retrieval, the correction halves Marepo's error (2.87 to 1.24 cm) but leaves ACE's essentially unchanged (1.05 to 1.01 cm). Table II extends the comparison to 12Scenes.

C. Uncertainty Calibration

Fig. 3 and Table III summarize coverage, AUSE and ECE over all 700 test queries for the LoFTR configuration (seed 0); the cascade of Table I is similar (coverage 59/75/83%, AUSE 0.063).

Three results stand out. First, the raw Laplace covariance covers essentially no queries at any level: a Gauss–Newton curvature estimate on heavy-tailed, spatially correlated photometric residuals is drastically over-confident, and a single temperature fitted on train-side frames recovers most of the diagonal. Second, the uncertainty is *informative*: sparsification by predicted uncertainty tracks the oracle (AUSE 0.063, Fig. 4a), so discarding the queries the model calls uncertain removes the ones that are wrong. Third, calibration comes from the same channel that drives accuracy: without the matching correction, coverage drops from 61/74/80 to 32/40/45% and ECE more than doubles (0.097 to 0.252), because bimodal errors cannot be described by one covariance scale.

Coverage is closest to nominal at the 68.3% level (61.1%) and falls short at higher levels (74.1 against 95.4%, 80.4 against 99.7%). The tempered d^2 median is 5.03 against the χ^2_6 ideal of 5.35, so the bulk is well scaled and the deficit lives in the tail (Fig. 3b). The gap is dominated by Stairs, where several poses of comparable evidence exist: the mixture (12) represents them, and the loss occurs in the moment matching that a single-Gaussian filter interface requires.

Comparison with a strong point estimator. We re-ran ACE [9] with the authors' pseudo-GT models and their `dsacstar` solver on the identical frames. It is level in accuracy and more reliable (pooled median 0.87 versus 0.89 cm; 3% versus 15% of queries beyond 5 cm), but its only confidence signal is the PnP inlier count; Marepo reports none, and the published DSAC* estimates [32] carry none, so a robot can only accept or reject them wholesale (dashed in Fig. 4b). Ranking by our tempered σ_t tracks the oracle (AUSE 0.063), whereas the inlier count barely orders ACE's errors (AUSE 0.166; rank correlation with true error 0.74 versus 0.50). Across deciles of reported confidence our median error spans 0.30 to 8.6 cm, a factor of 29, against 0.70 to 2.8 cm for the inlier count. Two consequences follow. On its most confident decile USGLoc is more than twice as accurate as ACE is on its own (0.30 versus 0.70 cm), so gating on our covariance isolates a subset on which we are the better method. And our failures are announced: the predicted σ_t on our >5 cm queries is 10–99$\times$ larger than on our successes on six scenes (median ratio), and only 1.3$\times$ on Stairs, where failures are confident registrations on the wrong tread.

What the uncertainty is worth downstream. Treating a translation error beyond 5 cm as a failure, σ_t separates failures from successes with a ROC AUC of 0.92 and recovers 95% of failures at a 17% false-positive rate. A robot that may decline

TABLE I

MEDIAN TRANSLATION/ROTATION ERROR (CM/°) ON 7SCENES UNDER THE SFM PSEUDO-GROUND TRUTH [32], AND SUCCESS RATE AT 5 CM/5° AND 2 CM/2° AVERAGED OVER SCENES; BEST IN BOLD, SECOND UNDERLINED. †: AS REPORTED BY THE AUTHORS. †r: RETRAINED UNDER THE SFM PSEUDO-GT, POSENET BY NEFES [35] AND MAREPO BY GS-CPR [12]. †s: MEDIANS AS PRINTED IN STDLOC [13], WHOSE SOURCES REPORT ONLY SUCCESS RATES. SUCCESS RATES ARE COMPUTED ON OUR 100 TEST QUERIES PER SCENE: DSAC* FROM THE AUTHORS' PUBLISHED ESTIMATES [32], ACE (WITH ITS DSACSTAR SOLVER) AND MAREPO FROM OUR RUNS OF THE RELEASED SFM PSEUDO-GT MODELS; GSPLATLOC^b: 5 CM/5° RATE AS REPORTED FOR ITS BASE SETTING (ITS FINE SETTING AND POSENET REPORT NO SUCCESS RATES); –: NOT AVAILABLE. m: OUR CORRECTION RUN AS A MAST3R [36] → DESCRIPTOR-FIELD CASCADE RATHER THAN LOFTR THROUGHOUT; SEED 0, 100 TEST QUERIES PER SCENE, ONE CONFIGURATION FOR EVERY SCENE.

	Median error (cm/°) ↓								Success (%) ↑	
Method	Chess	Fire	Heads	Office	Pumpkin	Kitchen	Stairs	Avg.	5 cm/5°	2 cm/2°
PoseNet [6]$^{\dagger r}$	10.00/4.02	27.00/10.00	18.00/13.00	17.00/5.97	19.00/4.67	22.00/5.91	35.00/10.50	21.00/7.74	–	–
Marepo [7]$^{\dagger r}$	1.90/0.83	2.30/0.92	2.10/1.24	2.90/0.93	2.50/0.88	2.90/0.98	5.90/1.48	2.90/1.04	83.43	34.29
DSAC* [8]$^{\dagger s}$	0.50/0.17	0.78/0.29	0.50/0.34	1.19/0.35	1.19/0.29	**0.72/0.21**	2.65/0.78	1.07/0.35	**97.71**	82.14
ACE [9]$^{\dagger s}$	0.55/0.18	0.83/0.33	0.53/0.33	1.05/**0.29**	1.06/**0.22**	0.77/**0.21**	2.89/0.81	1.10/0.34	97.14	82.71
GSplatLoc [11]† (Fine)	**0.39/0.13**	0.91/0.29	0.94/0.50	1.41/0.32	1.41/0.26	1.32/0.29	3.44/0.82	1.40/0.37	78.00^b	–
USGLoc (Ours, prior-free)^m	**0.39/0.13**	**0.51/0.22**	**0.48/0.31**	**1.04**/0.32	**1.02**/0.23	0.86/0.22	**1.43/0.43**	**0.82/0.27**	96.43	**87.14**

TABLE II

MEDIAN TRANSLATION/ROTATION ERROR (CM/°) ON 12SCENES UNDER THE SFM PSEUDO-GROUND TRUTH [32]; LOWER IS BETTER. §: AS PRINTED IN ULF-LOC [37]. ∗: COMPUTED BY US FROM THE AUTHORS' PUBLISHED DSAC* ESTIMATES [32]. ‖: OVERALL MEDIAN FROM THE 12SCENES STUDY OF [7], UNDER THE ORIGINAL D-SLAM POSES; PER-SCENE MEDIANS ARE NOT PUBLISHED (–). SUCCESS (%): RATE AT 5 CM/5° AND 2 CM/2° AVERAGED OVER SCENES, ON OUR 100 TEST QUERIES PER SCENE; DSAC* FROM ITS PUBLISHED ESTIMATES; –: NO PER-FRAME ESTIMATES PUBLISHED. OURS: TABLE I CONFIGURATION, SEED 0, 100 TEST QUERIES PER SCENE; –: SCENE STILL RUNNING.

	Apartment 1		Apartment 2				Office 1				Office 2			Success (%)	
Method	Kitchen	Living	Bed	Kitchen	Living	Luke	Gates362	Gates381	Lounge	Manolis	5a	5b	Avg.	5/5	2/2
PoseNet [6]$^{\|}$	–	–	–	–	–	–	–	–	–	–	–	–	9.40/3.90	–	–
Marepo [7]§	1.90/1.20	1.70/0.90	2.00/1.00	2.10/1.10	1.80/0.90	2.30/1.30	1.70/0.90	2.40/1.20	1.90/1.00	1.90/0.90	2.00/0.90	2.90/1.10	2.10/1.04	–	–
DSAC* [8]*	0.39/0.24	0.40/0.18	0.38/0.18	0.37/0.19	0.69/0.33	0.62/0.26	0.42/0.18	0.68/0.33	0.54/0.20	0.58/0.29	0.76/0.32	0.73/0.27	0.55/0.25	99.83	96.25
ACE [9]§	0.53/0.27	0.60/0.19	0.49/0.20	0.63/0.26	0.59/0.23	0.76/0.27	0.69/0.22	0.81/0.32	0.84/0.21	0.76/0.28	0.86/0.33	0.82/0.28	0.70/0.26	–	–
GSplatLoc [11]§	1.31/0.24	0.68/0.21	1.42/0.24	0.67/0.26	0.54/0.19	1.84/0.31	1.82/0.26	1.99/0.26	0.64/0.22	0.78/0.27	3.66/0.37	0.94/0.27	1.36/0.26	–	–
USGLoc (Ours)	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–

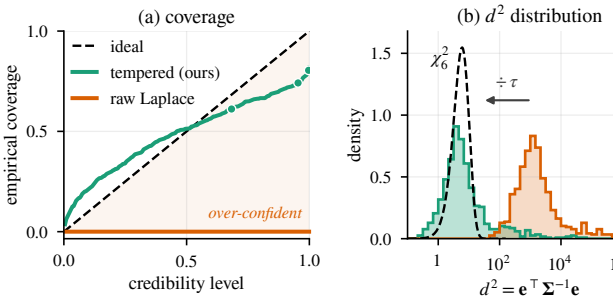


Fig. 3. Calibration on 7Scenes (700 test queries). (a) Empirical coverage vs. χ^2_6 credibility level; dots mark the levels of Table III. (b) Distribution of d^2: raw Laplace values are over two orders of magnitude too large; dividing by the held-out temperature τ moves them onto χ^2_6.

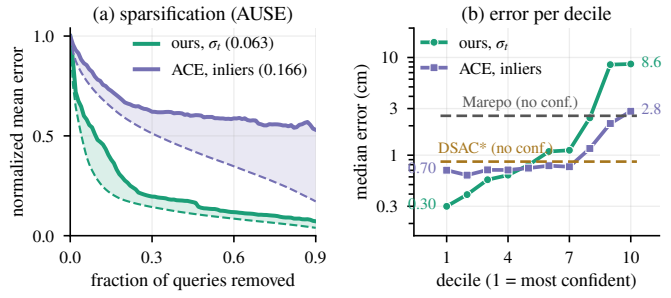


Fig. 4. Uncertainty quality against ACE [9] on the same 700 queries. (a) Sparsification by each method's own confidence, normalized by its own initial error; dashed: oracle (rank by true error); shaded: AUSE. (b) Median error per decile of reported confidence; DSAC* (published estimates) and Marepo report no confidence and are drawn at their overall median.

to answer can accept 81% of queries while keeping 95% of the accepted poses within 5 cm. Rejecting the least-confident half cuts our pooled median from 0.89 to 0.51 cm and the 90th percentile from 10.2 to 1.3 cm; the same rejection driven by ACE's inlier count moves its median only from 0.87 to 0.70 cm and its 90th percentile from 2.7 to 1.4 cm. For fusion the picture is mixed: in a constant-velocity EKF the covariance lowers trajectory error on Chess but raises it on Redkitchen and Stairs, where failures are confident, so a filter benefits only where the covariance separates failures sharply (supplementary material).

D. Robustness

We stress the Chess map, before the final fine-tune, by fitting from fewer frames (every 40th and 80th instead of every 10th) and by distilling semantics from fewer frames (every 80th and 200th instead of every 20th). The median moves from 0.56 to at most 0.78 cm, and the temperature keeps coverage at the 95.4% level between 83 and 89% in every case (79% for the full map): the uncertainty holds as the map gets worse.



Fig. 5. Map rendered at the pose returned by ACE, Marepo and USGLoc on identical queries, and a free-camera map view (□ ACE, ◊ Marepo, △ ours, ◯ true pose). Per scene, our largest margin over both baselines among sub-centimetre results, so favourable by design. Stamps: translation error; for USGLoc also the percentile of its σ_t.

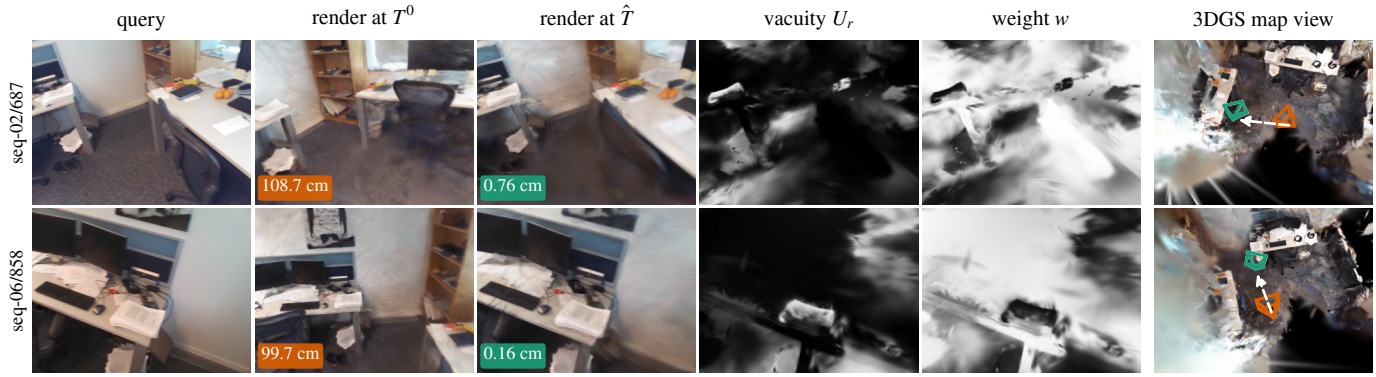


Fig. 6. The stages on two Office test queries. Left to right: query, render at the retrieved pose T^0 and at the refined pose $\hat{T}$ (stamps: translation error), rendered vacuity U_r (bright = vacuous), weight $w = e^{-\gamma U_r}$, and a map view with T^0 (orange), $\hat{T}$ (green) and the true pose (white). Rows: final error below 1 cm with the largest coarse error, one per sequence, so favourable by design.

TABLE III
CALIBRATION ON 7SCENES, POOLED OVER ALL 700 TEST QUERIES (TEMPERED). COVERAGE SHOULD MATCH THE NOMINAL LEVEL; AUSE/ECE LOWER IS BETTER. THE RAW (UNTEMPERED) LAPLACE COVARIANCE IS SEVERELY OVERCONFIDENT.

Method	Cov. 68 (%)	Cov. 95 (%)	Cov. 99.7 (%)	AUSE↓ (t)	ECE↓
Ideal χ^2 ref.	68.3	95.4	99.7	–	–
Raw Laplace (no τ)	0.0	0.0	0.0	0.061	0.505
– matching correction	31.6	39.9	44.6	0.112	0.252
USGLoc (Ours)	**61.1**	**74.1**	**80.4**	**0.063**	**0.097**

E. Ablation Study

Table IV removes one component at a time on the easiest and the hardest scene. Stairs errors are bimodal (about 60% of queries near 1.8 cm, most of the rest beyond 25 cm), so its median must be read next to the success rate. The matching correction dominates: without it the medians are 13.6 and 45.1 cm. Among the rest, the components that act after the correction matter most: on Stairs the pixel weight of (10) is worth 28% (3.93 to 2.83 cm), the inlier-based mixture weights of (13) 2.1× over retrieval-prior weights, and reporting the mode rather than the mixture mean 4×. Keeping $K{=}3$ hypotheses costs nothing on Chess but makes the uncertainty informative: with $K{=}1$ the AUSE degrades from 0.056 to 0.373 on Chess and from 0.291 to 0.656 on Stairs, because one

TABLE IV
ABLATION ON CHESS AND STAIRS (SEED 0, 100 QUERIES): MEDIAN TRANSLATION ERROR (CM), SUCCESS RATE AT 5 CM/5° (%), AND TRANSLATION AUSE. INDENTED ROWS REMOVE OR REPLACE ONE COMPONENT OF *Full (ablation maps)*, THE METHOD ON THE MAPS USED BEFORE THE FINAL FINE-TUNE.

	Chess			Stairs		
Variant	med.	SR	AUSE	med.	SR	AUSE
USGLoc (final maps)	0.47	89	0.058	2.40	56	0.297
Full (ablation maps)	0.56	88	0.056	2.83	58	0.291
− matching correction (8)	13.55	39	0.068	45.08	6	0.220
− correspondence gate	0.54	88	0.065	2.90	57	0.262
− instance gate (λ_u=0)	0.56	88	0.059	2.90	58	0.278
− pixel weight (γ=0)	0.55	86	0.045	3.93	52	0.249
Dempster (no conflict)	0.55	88	0.056	2.79	58	0.287
retrieval-prior weights (pre-(13))	0.57	87	0.044	5.92	49	0.293
mixture mean instead of mode	0.55	87	0.052	11.30	30	0.184
point estimate (K=1)	0.56	87	0.373	4.04	51	0.656
snap only (no refinement)	0.62	92	0.077	3.54	54	0.229
softmax head, matched map	0.62	87	0.061	11.60	44	0.297

Laplace covariance cannot represent an ambiguous scene. By contrast, the correspondence and instance gates are within run noise once the correction is in place (Stairs median ±0.07 cm), and Dempster and Yager fusion are indistinguishable: the evidential channel earns its keep through the pixel weight and the mixture. Skipping refinement (snap only) trades 0.06–0.7 cm of accuracy for better coverage on Stairs (54/57 against 38/47%). On matched maps the softmax head is worse on both scenes (0.62 against 0.56 cm on Chess; on Stairs failures beyond 5 cm rise from 42 to 56%), with higher AUSE.

F. Limitations

First, each hypothesis's covariance is a unimodal local Laplace approximation: the mixture captures ambiguity across hypotheses, but heavy-tailed error within a basin is under-modeled, and coverage stays below nominal in the tails, most on Stairs. Second, the evidential state inherits its frozen front end: content FastSAM does not segment or CLIP cannot discriminate earns vacuity rather than evidence. Third, accuracy is bounded by the map and the matcher: USGLoc is not the most accurate localizer, ACE still fails slightly less often (2.9% versus 3.6% of queries beyond 5 cm), and our remaining deficit is Kitchen, where both scene-coordinate regressors are more accurate, and the Pumpkin failure tail. Fourth, conflict handling is rule-bound (neither Dempster nor Yager models disagreement over time), and the evaluation is indoor and per-scene. The contribution is the calibrated distribution and the quality of its uncertainty, not state-of-the-art point accuracy.

G. Runtime

On the RTX 5070 Ti the median query takes 0.30 s to encode (FastSAM + CLIP) and 6.3 s to localize (per scene 5.3–7.0 s). The correction costs about 0.25 s per hypothesis (2 s for the top-8); the rest is the per-hypothesis refinement and its Laplace Jacobian. With a single hypothesis (K=1) the median falls from 6.0 to 5.9 s on Chess and from 6.9 to 4.0 s on Stairs. Map training takes 60–95 s per scene at 320×240 and 140–370 s at 640×480.

V. CONCLUSION

We presented USGLoc, a semantic 3DGS localizer that returns a calibrated distribution over camera pose rather than a point. One evidential state, created when observations are encoded and never discarded, gates instance matching and the correspondences of the matching correction, weights the refinement residual per pixel, and scales the Laplace covariance through a held-out temperature, yielding a Gaussian-mixture posterior on $SE(3)$ that condenses to one Gaussian for filtering. Prior-free on 7Scenes, USGLoc reaches 0.82 cm/0.27° on average, the lowest in Table I, with a covariance whose ordering of its own errors approaches the oracle; it is the only method in the table that reports one. Future work includes an open-set semantic extension, tighter coupling with the navigation filter, learned covariance heads, and on-board experiments on an indoor MAV.

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